UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
18/564,867	11/28/2023	Motohito TAKEZAKI	CRI-54831	5205

97379 7590 10/30/2025

Rankin, Hill & Clark LLP
23755 Lorain Road, Suite 200
North Olmsted, OH 44070

EXAMINER	
PEACE, RHONDA S	

ART UNIT	PAPER NUMBER
2874	

NOTIFICATION DATE	DELIVERY MODE
10/30/2025	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

ishihara@rankinhill.com
millar@rankinhill.com

Office Action Summary	Application No. 18/564,867	Applicant(s) TAKEZAKI, Motohito	
	Examiner R. PEACE	Art Unit 2874	AIA (First Inventor to File) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☐ Responsive to communication(s) filed on _____.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☐ This action is **FINAL.** 2b) ☑ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☑ Claim(s) 1-23 is/are pending in the application.

5a) Of the above claim(s) _____ is/are withdrawn from consideration.

6) ☐ Claim(s) _____ is/are allowed.

7) ☑ Claim(s) 1-7 and 9-23 is/are rejected.

8) ☑ Claim(s) 8 is/are objected to.

9) ☐ Claim(s) _____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

Application Papers

10)☐ The specification is objected to by the Examiner.

11)☑ The drawing(s) filed on 11/28/23 is/are: a)☑ accepted or b)☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☑ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a)☑ All b)☐ Some** c)☐ None of the:

1.☑ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☑ Notice of References Cited (PTO-892)

2) ☑ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date 11/28/23.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date _____.

4) ☐ Other: _____.

DETAILED ACTION

Notice of Pre-AIA or AIA Status

The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

Priority

Receipt is acknowledged of certified copies of papers required by 37 CFR 1.55.

Information Disclosure Statement

The information disclosure statement (IDS) submitted on 11/28/23 was filed in compliance with the provisions of 37 CFR 1.97. Accordingly, the information disclosure statement is being considered by the examiner.

Claim Rejections - 35 USC § 102

In the event the determination of the status of the application as subject to AIA 35 U.S.C. 102 and 103 (or as subject to pre-AIA 35 U.S.C. 102 and 103) is incorrect, any correction of the statutory basis (i.e., changing from AIA to pre-AIA) for the rejection will not be considered a new ground of rejection if the prior art relied upon, and the rationale supporting the rejection, would be the same under either status.

The following is a quotation of the appropriate paragraphs of 35 U.S.C. 102 that form the basis for the rejections under this section made in this Office action:

> A person shall be entitled to a patent unless –
>
> (a)(1) the claimed invention was patented, described in a printed publication, or in public use, on sale, or otherwise available to the public before the effective filing date of the claimed invention.

Claim(s) 1, 4, 7, 11, and 12 is/are rejected under 35 U.S.C. 102(a)(1) as being anticipated by Lange et al. (US 2005/0069265 A1), cited in the IDS filed 11/28/23.

Re. Claim 1, Lange et al. discloses an optical fiber assembly, comprising:

a plate-like ferrule 11 that is formed of ceramics, and has optical fiber insertion holes 20 and guide holes 22 for insertion of a guide pin 62 (Fig. 2A and 6A; [0030]-[0031] and [0040]); and optical fibers 50 inserted into the optical fiber insertion holes 20 of the ferrule 11 (Figs. 5A-5E; [0035], [0039]), wherein the optical fibers are fixed to the optical fiber insertion holes of the ferrule with an adhesive filled in the optical fiber insertion holes 20 (Figs. 2D and 5E; [0037]). Due to the tapered nature of holes 20 (Fig. 2D; [0031]), adhesive applied to the cavity 54 inherently fills at least a portion of the holes 20.

Re. Claim 4, Lange et al. discloses the ferrule 11 is a multi-fiber optical connector ferrule that has at least 12 optical fiber insertion holes 20 (Fig. 2A).

Re. Claim 7, Lange et al. discloses the ferrule includes a thin portion 11 where the plate main body formed to have the optical fiber insertion holes 20 has a small thickness, and a thick portion 52, and guide grooves that communicates with the optical fiber insertion holes 20 are formed in an upper end face of the thick portion 52 (Figs. 5A-5E; [0035]).

Re. Claim 11, Lange et al. discloses the plate main body has a thickness t ranging from 0.3 to 3.0 mm, inclusive (Fig. 2D; [0031]).

Re. Claim 12, Lange et al. discloses an optical module, comprising: a substrate 3; a photoelectric conversion element 4 mounted on the substrate 3; and the optical fiber assembly discussed above that is mounted close to the photoelectric conversion element 4 or a silicon optical waveguide, and optically connected (Figs. 6A and 7B; [0040]-[0042]).

Claim Rejections - 35 USC § 103

In the event the determination of the status of the application as subject to AIA 35 U.S.C. 102 and 103 (or as subject to pre-AIA 35 U.S.C. 102 and 103) is incorrect, any correction of the statutory basis (i.e., changing from AIA to pre-AIA) for the rejection will not be considered a new ground of rejection if the prior art relied upon, and the rationale supporting the rejection, would be the same under either status.

The following is a quotation of 35 U.S.C. 103 which forms the basis for all obviousness rejections set forth in this Office action:

> A patent for a claimed invention may not be obtained, notwithstanding that the claimed invention is not identically disclosed as set forth in section 102, if the differences between the claimed invention and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. Patentability shall not be negated by the manner in which the invention was made.

The factual inquiries for establishing a background for determining obviousness under 35 U.S.C. 103 are summarized as follows:

1. Determining the scope and contents of the prior art.

2. Ascertaining the differences between the prior art and the claims at issue.

3. Resolving the level of ordinary skill in the pertinent art.

4. Considering objective evidence present in the application indicating obviousness or nonobviousness.

Claim(s) 2, 3, 5, 6, 9, 10, 13-23 is/are rejected under 35 U.S.C. 103 as being unpatentable over Lange et al. (US 2005/0069265 A1).

Re. Claim 2, Lange et al. discloses the optical fiber assembly as discussed above, but fails to explicitly disclose the ceramics includes machinable *ceramics*, and the optical fiber contains *quartz glass*.

Each of the claimed materials are well known in the art for their claimed purposes, and their use would have been obvious to one of ordinary skill in the art before the effective filing date of the claimed invention. It has been held to be within the general skill of a worker in the art to select a known material on the basis of its suitability for the intended use as a matter of obvious design choice. *In re Leshin*, 125 USPQ 416.

Re. Claim 3, Lange et al. discloses the optical fiber assembly as discussed above, but fails to explicitly disclose the ceramics has a hardness *lower* than a hardness of the optical fiber.

The claimed arrangement would have been obvious to one of ordinary skill in the art before the effective filing date of the claimed invention for the purpose of minimizing damage to the optical fiber during the insertion process. It has been held to be within the general skill of a worker in the art to select a known material on the basis of its suitability for the intended use as a matter of obvious design choice. *In re Leshin*, 125 USPQ 416.

Re. Claim 5, Lange et al. discloses the optical fiber assembly as discussed above, but fails to explicitly disclose an amount of dimensional variation of the ferrule heated at 260°C is *no more than 0.5 μm*.

The claimed arrangement would have been obvious to one of ordinary skill in the art before the effective filing date of the claimed invention for the purpose of minimizing deformation during the heating process. It has been held to be within the general skill of a worker in the art to select a known material on the basis of its suitability for the intended use as a matter of obvious design choice. *In re Leshin*, 125 USPQ 416.

Re. Claim 6, Lange et al. discloses the optical fiber assembly as discussed above, wherein the ferrule includes a plate main body formed to have a rectangular shape (Fig. 2A), but fails to disclose a *surface identification structure* is formed at one corner of the plate main body.

Surface identification structures are well known in the art, and the claimed arrangement would have been obvious to one of ordinary skill in the art before the effective filing date of the claimed invention for the purpose of ensuring the plate is coupled to the remaining structure in the desired manner. "A person of ordinary skill is also a person of ordinary creativity, not an automaton" – '[w]hen there is a design need or market pressure to solve a problem and there are a finite number of identified, predictable solutions, a person of ordinary skill has good reason to pursue the known options within his or her technical grasp. If this leads to the anticipated success, it is likely the product not of innovation but of ordinary skill and common sense." *KSR International Co. v. Teleflex Inc.*, 550 USPQ2d 1385 (2007).

Re. Claim 9, Lange et al. renders obvious the plate as discussed above.

Lange et al. fails to disclose an arrangement wherein *an adhesive reservoir including the optical fiber insertion holes is provided in a surface of the plate main body in a concave manner*.

Adhesive reservoirs are well known in the optical connector arts, and the claimed arrangement would have been obvious to one of ordinary skill before the effective filing date of the claimed invention for the purpose of reliably attaching the optical fibers to the optical fiber insertion holes. "A person of ordinary skill is also a person of ordinary creativity, not an automaton" – '[w]hen there is a design need or market pressure to

solve a problem and there are a finite number of identified, predictable solutions, a person of ordinary skill has good reason to pursue the known options within his or her technical grasp. If this leads to the anticipated success, it is likely the product not of innovation but of ordinary skill and common sense." *KSR International Co. v. Teleflex Inc.*, 550 USPQ2d 1385 (2007).

Re. Claim 10, Lange et al. renders obvious the plate in Fig 2A as discussed above.

Lange et al. fails to disclose an arrangement wherein the guide hole is formed to have *a hat-shaped cross section, and has a small hole, and a large hole formed around the small hole*.

In a separate embodiment, Lange et al. illustrates a guide hole 24 is formed to have a hat-shaped cross section, and has a small hole, and a large hole formed around the small hole (Fig. 2B; [0031]).

The claimed arrangement would have been obvious to one of ordinary skill before the effective filing date of the claimed invention for the purpose of providing the desired ferrule. "A person of ordinary skill is also a person of ordinary creativity, not an automaton" – '[w]hen there is a design need or market pressure to solve a problem and there are a finite number of identified, predictable solutions, a person of ordinary skill has good reason to pursue the known options within his or her technical grasp. If this leads to the anticipated success, it is likely the product not of innovation but of ordinary skill and common sense." *KSR International Co. v. Teleflex Inc.*, 550 USPQ2d 1385 (2007).

Re. Claims 13-14, 16, 18, 21, and 23, Lange et al. discloses the optical fiber assembly and optical module as discussed above, comprising a plate 11 and having at least twelve optical fiber insertion holes 20; and guide holes 22 for guide pin insertion (Fig. 2A and 6A; [0030]-[0031] and [0040]).

However, Lange et al. does not disclose the plate having a hardness of *no more than 8.0 GPa*.

The claimed arrangement would have been obvious to one of ordinary skill in the art before the effective filing date of the claimed invention for the purpose of minimizing damage to the optical fiber during the insertion process. It has been held to be within the general skill of a worker in the art to select a known material on the basis of its suitability for the intended use as a matter of obvious design choice. *In re Leshin*, 125 USPQ 416.

Re. Claim 15, Lange et al. renders obvious the plate as discussed above.

However, Lange et al. fails to explicitly disclose an amount of dimensional variation of the ferrule heated at 260°C is *no more than 0.5 μm*.

The claimed arrangement would have been obvious to one of ordinary skill in the art before the effective filing date of the claimed invention for the purpose of minimizing deformation during the heating process. It has been held to be within the general skill of a worker in the art to select a known material on the basis of its suitability for the intended use as a matter of obvious design choice. *In re Leshin*, 125 USPQ 416.

Re. Claim 17, Lange et al. discloses the optical fiber assembly as discussed above, wherein the ferrule includes a plate main body formed to have a rectangular

shape (Fig. 2A), but fails to disclose the *machined portion is a surface identification structure formed at a corner of the plate main body*.

Surface identification structures are well known in the art, and the claimed arrangement would have been obvious to one of ordinary skill in the art before the effective filing date of the claimed invention for the purpose of ensuring the plate is coupled to the remaining structure in the desired manner. "A person of ordinary skill is also a person of ordinary creativity, not an automaton" – '[w]hen there is a design need or market pressure to solve a problem and there are a finite number of identified, predictable solutions, a person of ordinary skill has good reason to pursue the known options within his or her technical grasp. If this leads to the anticipated success, it is likely the product not of innovation but of ordinary skill and common sense." *KSR International Co. v. Teleflex Inc.*, 550 USPQ2d 1385 (2007).

Re. Claim 19, Lange et al. renders obvious the plate as discussed above.

However, Lange et al. fails to explicitly disclose *the machined portion includes an adhesive reservoir formed in a surface of the plate main body having the optical fiber insertion holes*.

Adhesive reservoirs are well known in the optical connector arts, and the claimed arrangement would have been obvious to one of ordinary skill before the effective filing date of the claimed invention for the purpose of reliably attaching the optical fibers to the optical fiber insertion holes. "A person of ordinary skill is also a person of ordinary creativity, not an automaton" – '[w]hen there is a design need or market pressure to solve a problem and there are a finite number of identified, predictable solutions, a person of ordinary skill has good reason to pursue the known options within his or her

technical grasp. If this leads to the anticipated success, it is likely the product not of innovation but of ordinary skill and common sense." *KSR International Co. v. Teleflex Inc.*, 550 USPQ2d 1385 (2007).

Re. Claim 20, Lange et al. renders obvious the plate in Fig 2A as discussed above.

Lange et al. fails to disclose an arrangement wherein the guide hole is formed to have *a hat-shaped cross section, and has a small hole, and a large hole formed around the small hole*.

In a separate embodiment, Lange et al. illustrates a guide hole 24 is formed to have a hat-shaped cross section, and has a small hole, and a large hole formed around the small hole (Fig. 2B; [0031]).

The claimed arrangement would have been obvious to one of ordinary skill before the effective filing date of the claimed invention for the purpose of providing the desired ferrule. "A person of ordinary skill is also a person of ordinary creativity, not an automaton" – '[w]hen there is a design need or market pressure to solve a problem and there are a finite number of identified, predictable solutions, a person of ordinary skill has good reason to pursue the known options within his or her technical grasp. If this leads to the anticipated success, it is likely the product not of innovation but of ordinary skill and common sense." *KSR International Co. v. Teleflex Inc.*, 550 USPQ2d 1385 (2007).

Re. Claim 22, Lange et al. discloses the optical fiber assembly as discussed above, but fails to explicitly disclose *an optical connector is connected to another end of*

the optical fiber, and the optical connector is an MT connector that includes an MT ferrule.

Terminating optical fibers with optical connectors is well known in the art, as are MT ferrules, and the use of the same as claimed would have been obvious to one of ordinary skill in the art before the effective filing date of the claimed invention for the purpose of connecting the optical fiber with another optical system using conventional means for the use and processing of optical signals. "A person of ordinary skill is also a person of ordinary creativity, not an automaton" – '[w]hen there is a design need or market pressure to solve a problem and there are a finite number of identified, predictable solutions, a person of ordinary skill has good reason to pursue the known options within his or her technical grasp. If this leads to the anticipated success, it is likely the product not of innovation but of ordinary skill and common sense." *KSR International Co. v. Teleflex Inc.*, 550 USPQ2d 1385 (2007).

Allowable Subject Matter

Claim 8 is objected to as being dependent upon a rejected base claim, but would be allowable if rewritten in independent form including all of the limitations of the base claim and any intervening claims.

The following is a statement of reasons for the indication of allowable subject matter:

Re. Claim 8, the prior art does not disclose or reasonably suggest the claimed invention, specifically wherein at least two rows of the optical fiber insertion holes are arranged*, the thin portion has at least two thicknesses, and the guide grooves are*

provided stepwise on each row of the optical fiber insertion holes, in combination with the remaining limitations of the claims.

The most applicable prior art, Lange et al. (US 2005/0069265 A1), addressed above, fails to disclose or reasonably suggest the claimed invention, specifically those portions *highlighted* above in combination with the remaining limitations of the claims.

Conclusion

The prior art made of record and not relied upon is considered pertinent to applicant's disclosure. See the attached PTO_892.

Any inquiry concerning this communication or earlier communications from the examiner should be directed to R. PEACE whose telephone number is (571)272-8580. The examiner can normally be reached 9-5 pm.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at http://www.uspto.gov/interviewpractice.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Uyen-Chau Le can be reached at (571) 272-2397. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of published or unpublished applications may be obtained from Patent Center. Unpublished application information in Patent Center is available to registered users. To file and manage patent submissions in Patent Center, visit: https://patentcenter.uspto.gov. Visit https://www.uspto.gov/patents/apply/patent-center for more information about Patent Center and

https://www.uspto.gov/patents/docx for information about filing in DOCX format. For additional questions, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/RHONDA S PEACE/
Primary Examiner, Art Unit 2874
10/28/25